Digital Signal Processing

Spectral Estimation Techniques

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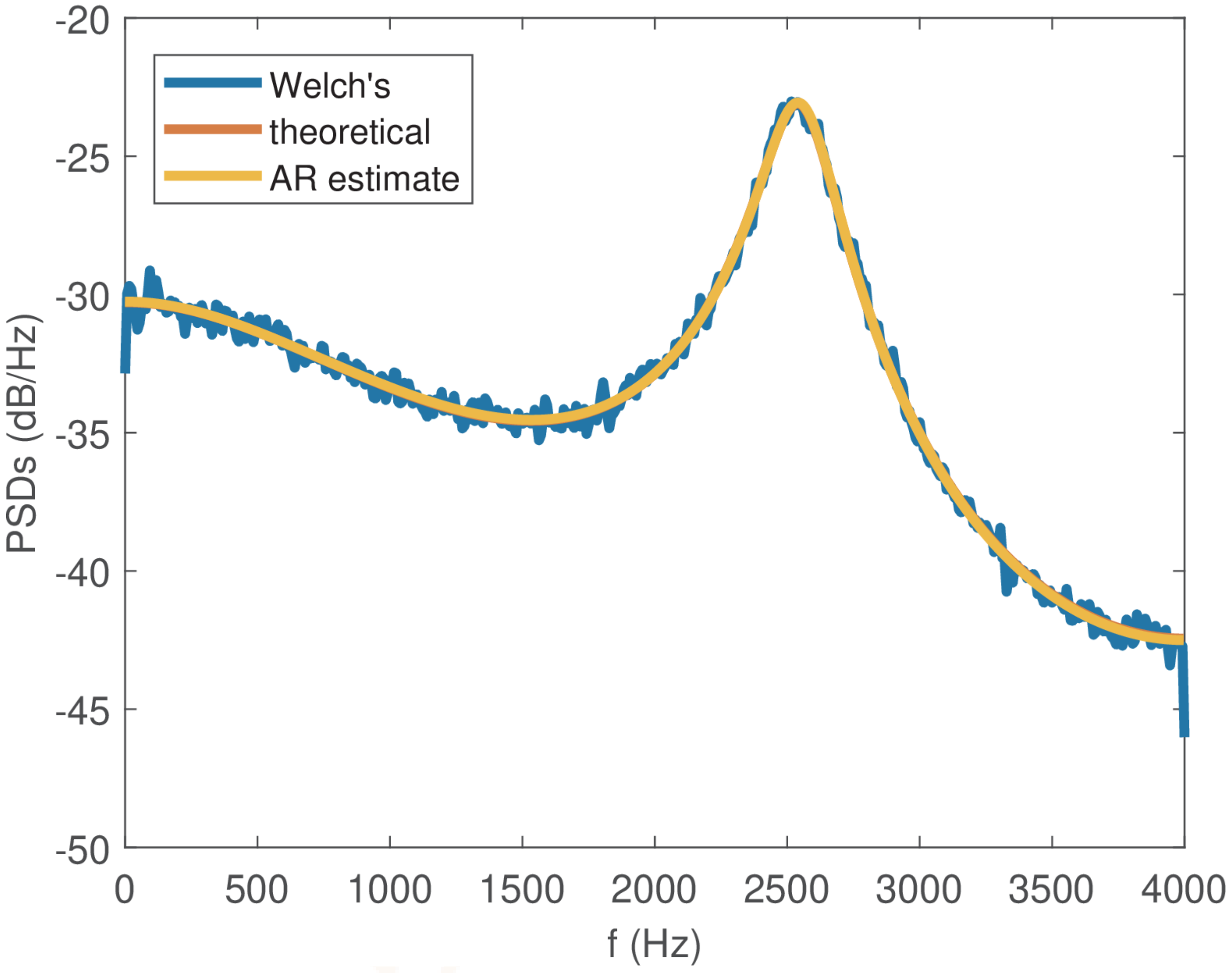
This chapter's Goal

Estimate how the signal power is distributed over frequency, which is different from frequency estimation

Learn about the tools used for spectral estimation

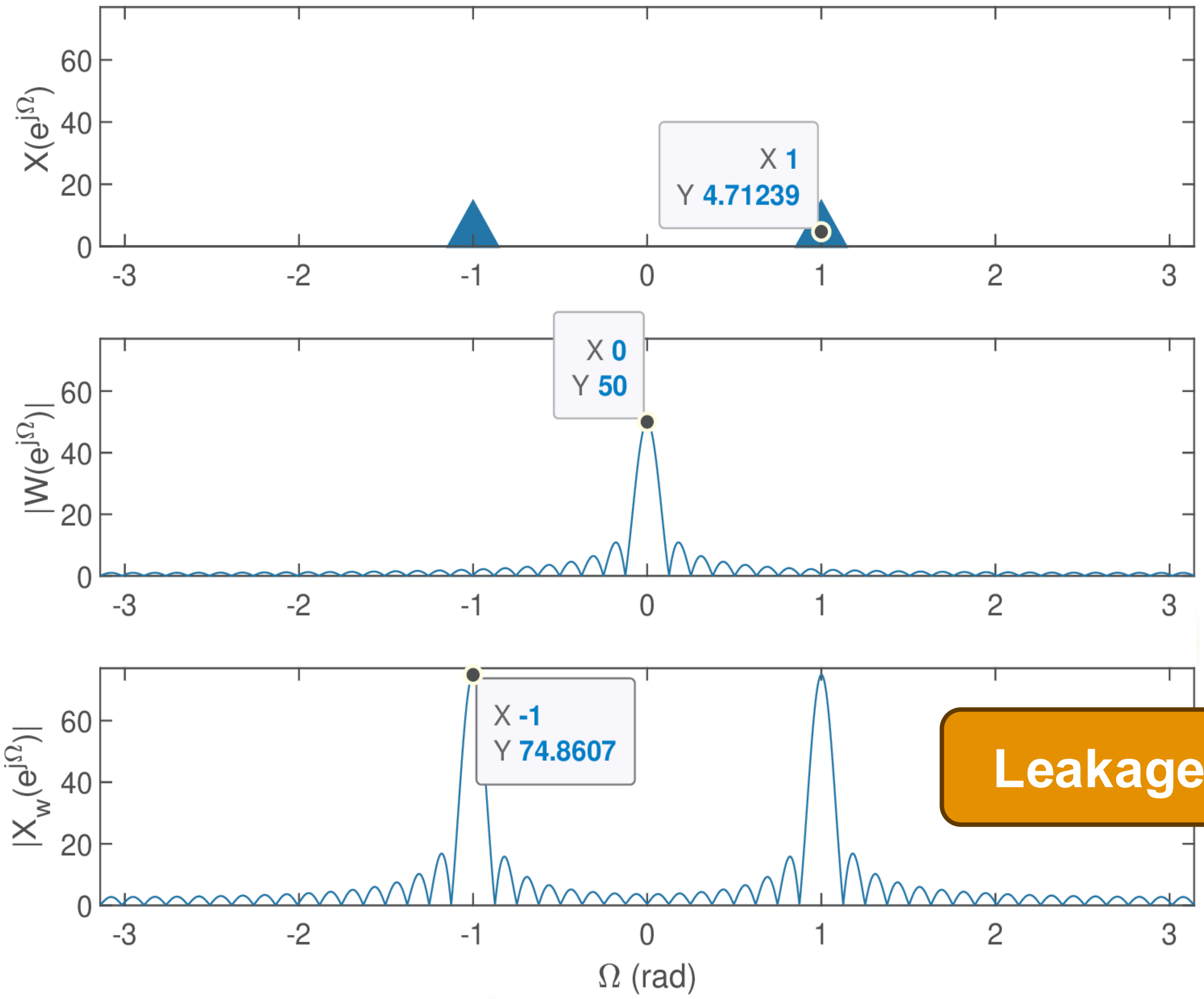


Windows for Spectral Analysis

As discussed previously, collected samples are finite

The act of collecting samples in finite memory computers is treated as windowing

$$x_w[n] = x[n]w[n]$$
$$\Leftrightarrow$$
$$X_w(e^{j\Omega}) = X(e^{j\Omega}) \circledast W(e^{j\Omega})$$

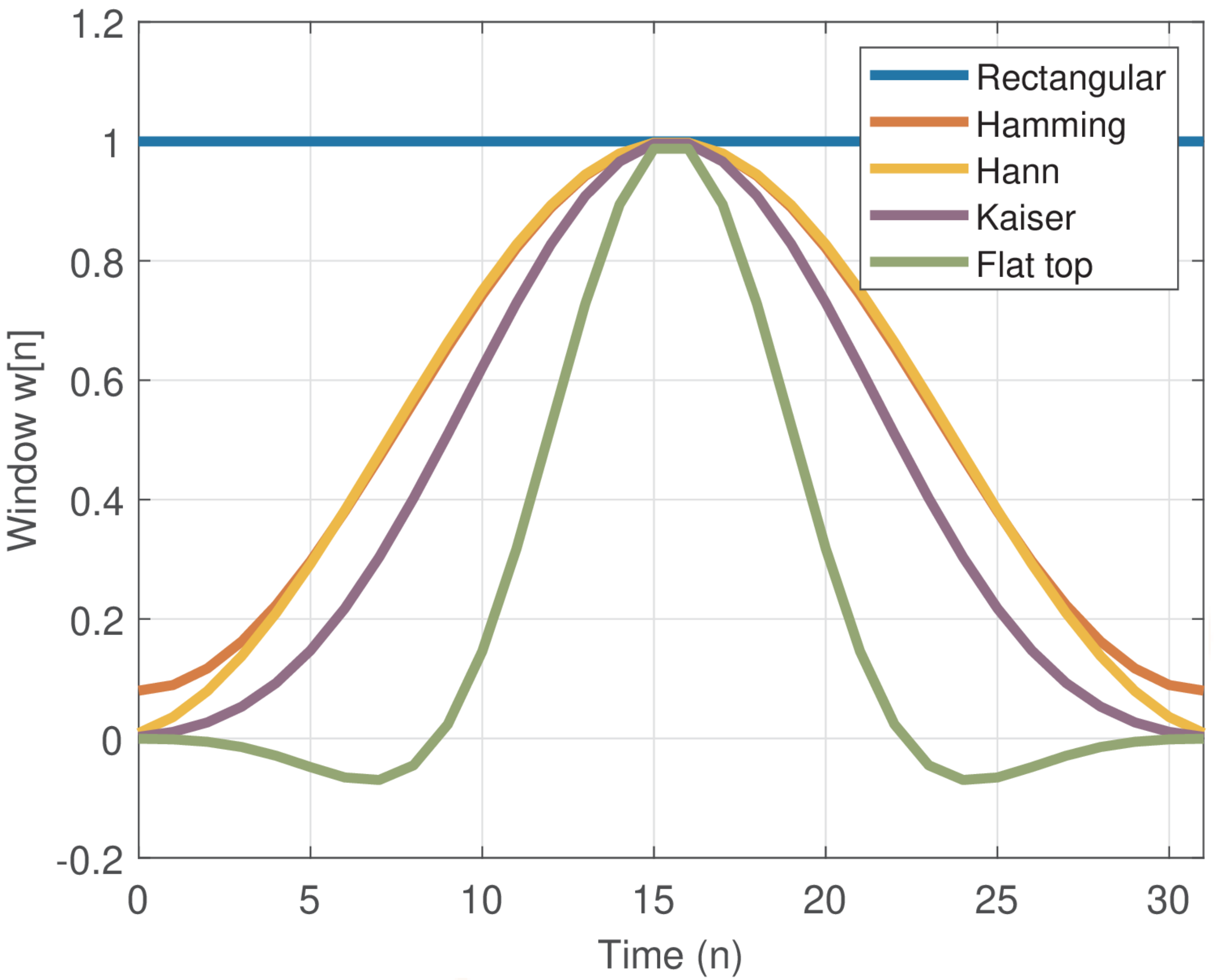


Leakage

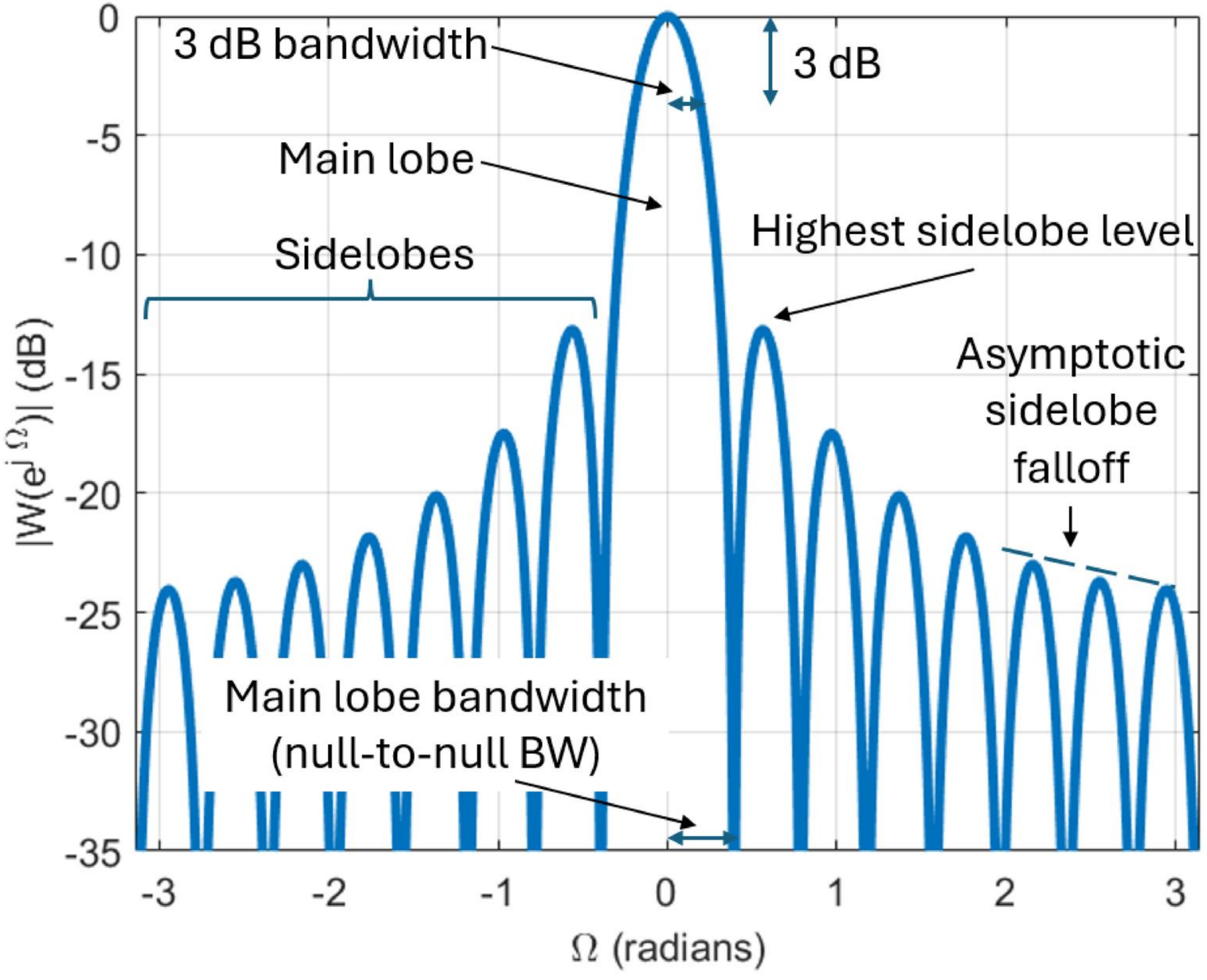
Commonly used windows

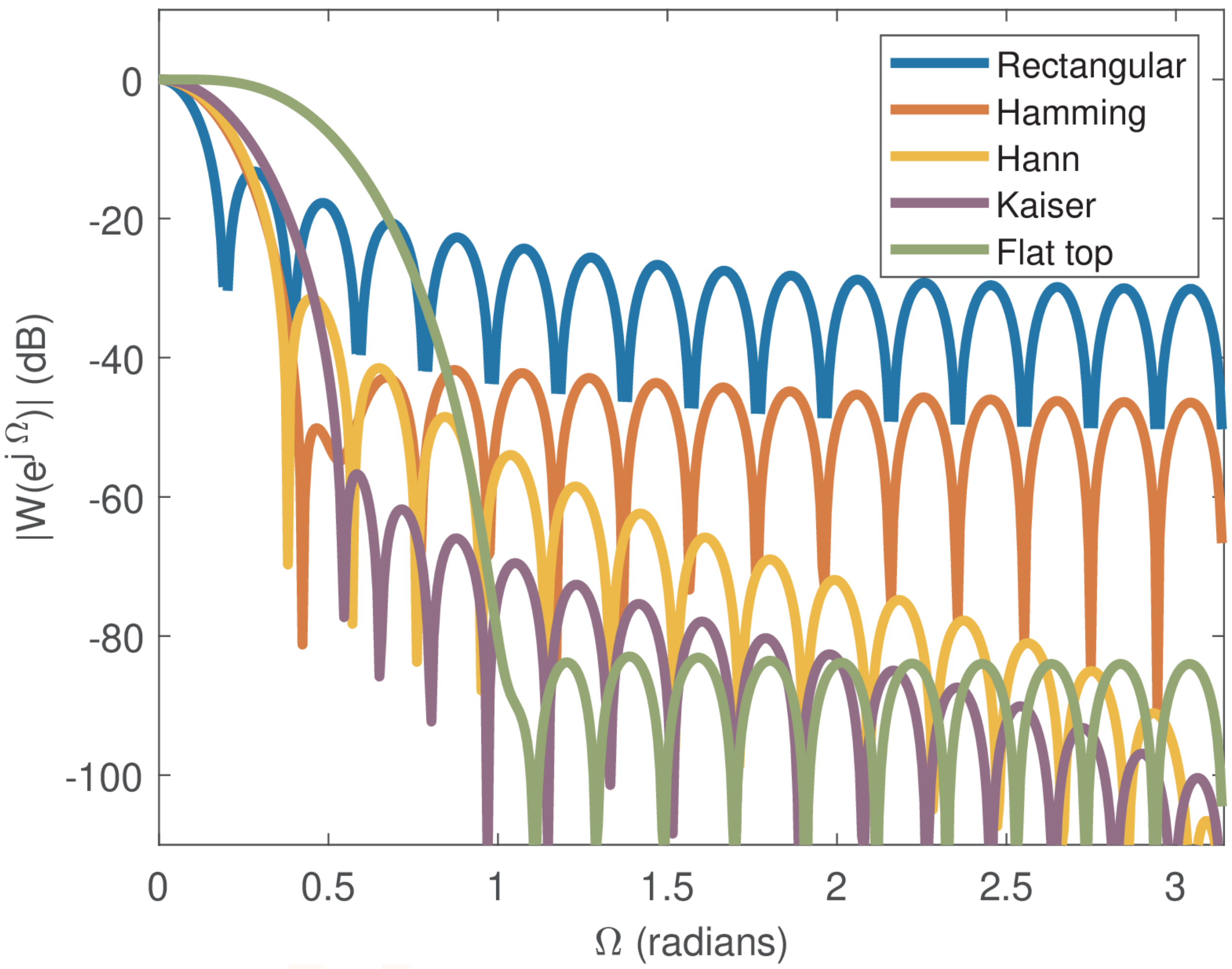
Rectangular	`numpy.ones`
Hamming	`numpy.hamming`
Hann	`numpy.hanning`
Kaiser	`numpy.kaiser`
Flat top	`scipy.signal.windows.flattop`

Others: Blackman-Harris, Chebyshev, etc.



Commonly used windows - Spectra

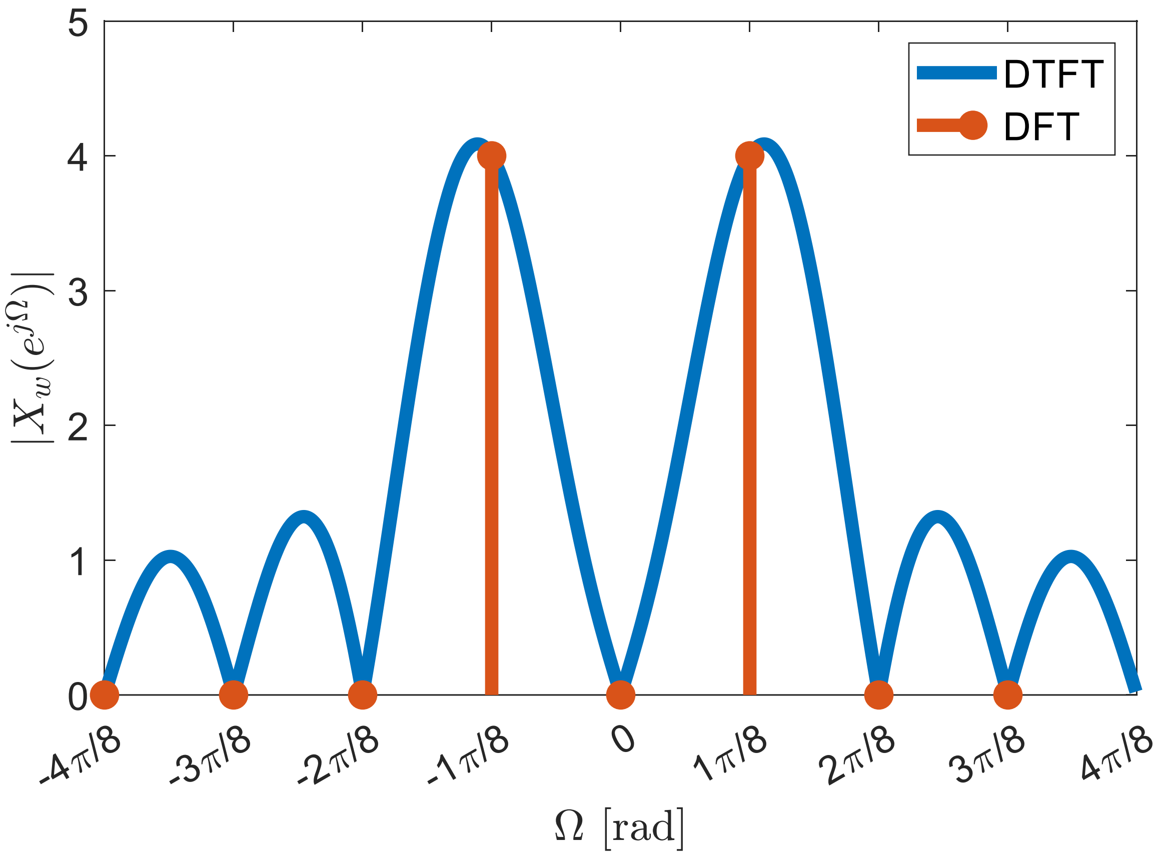




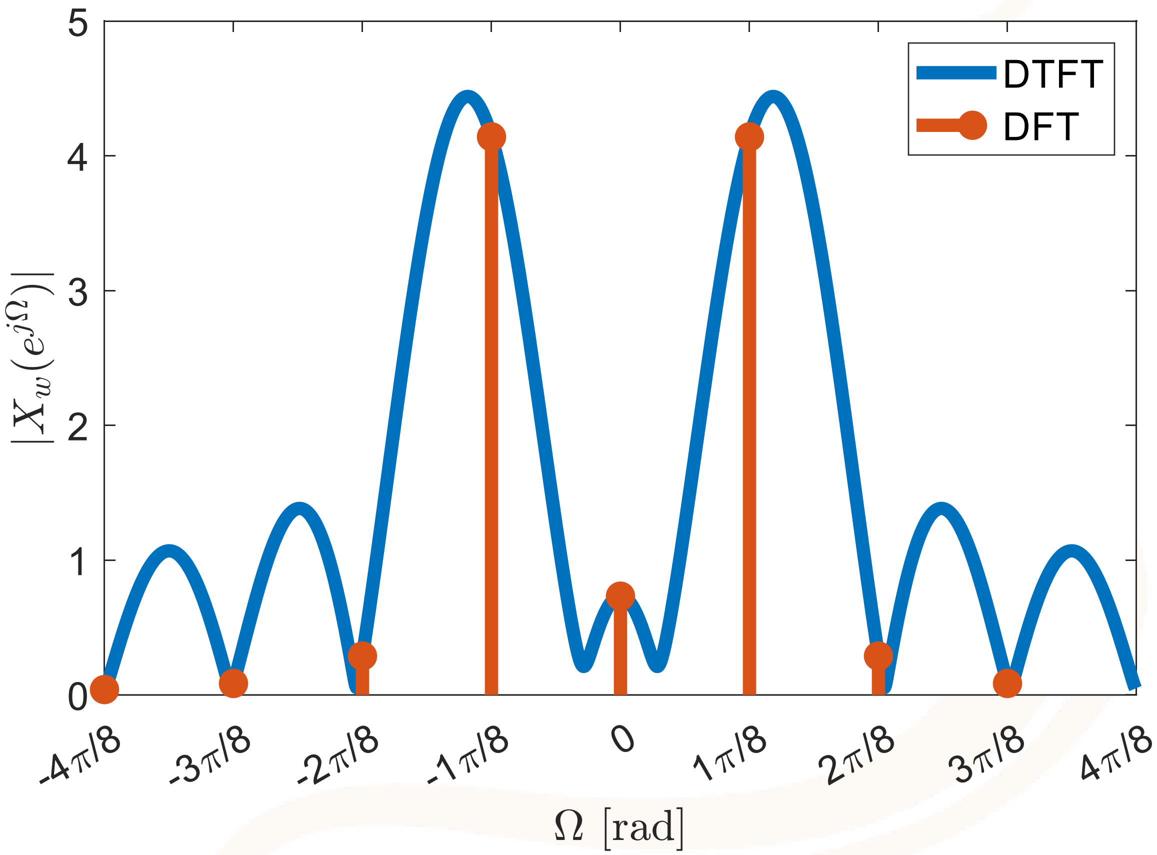
Picket-fence

Since DFT is discrete, we see the DTFT "through a fence"

Ex: Rect window and $N = 8$



$x[n] = \cos(1(2\pi/8)n)$



$x[n] = \cos(1.1(2\pi/8)n)$

Detecting Bin-centered vs not bin-centered sinusoids

A sinusoid is bin-centered if $\frac{\Omega}{2\pi} = \frac{m}{N}$, where N is the number of bins

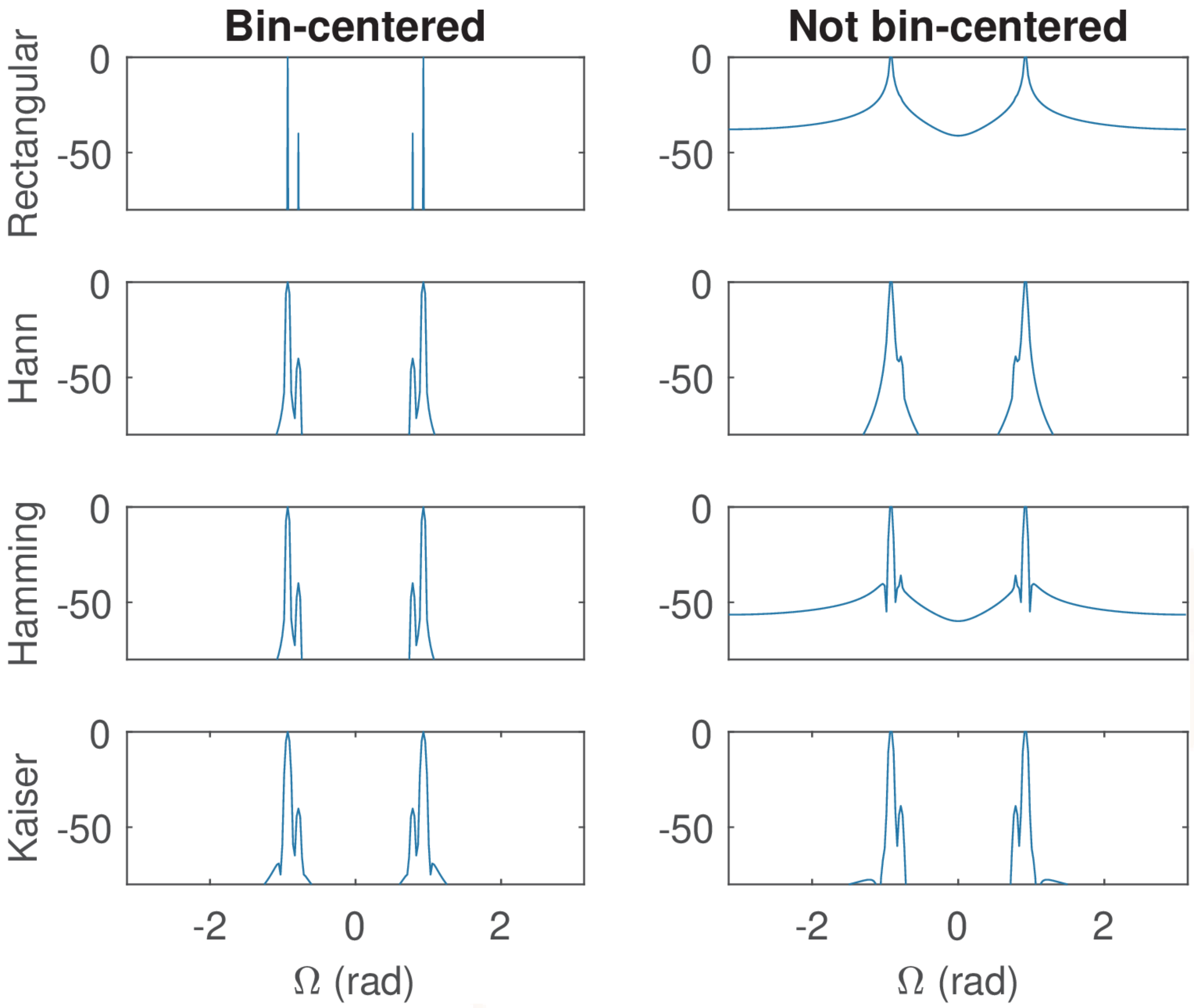
N=256

Bin-centered

$$x_1[n] = 1\cos\left((64\pi/256)n + \pi/3\right) + 100\,cos\left((76\pi/256)n + \pi/3\right)$$

Not bin-centered

$$x_2[n] = 1\cos\left((64\pi/256)n + \pi/3\right) + 100\,cos\left((75\pi/256)n + \pi/3\right)$$



ESD

Deterministic, finite energy signals

Time	Ind. var.	ESD definition	ESD main property
Continuous-time	f (Hz)	$G(f) = \|X(f)\|^2$	$E = \int_{-\infty}^{\infty} G(f)\mathrm{d}f$
Continuous-time	ω (rad/s)	$G(\omega) = \|X(\omega)\|^2$	$E = \frac{1}{2\pi}\int_{-\infty}^{\infty} G(\omega)\mathrm{d}\omega$
Discrete-time	Ω (rad)	$G(e^{j\Omega}) = \|X(e^{j\Omega})\|^2$	$E = \frac{1}{2\pi}\int_{<2\pi>} G(e^{j\Omega})\mathrm{d}\Omega$

Let $X(e^{j\Omega}) = 20, |\Omega| < 0.4\pi$

If F_s is 100Hz, the energy in the bands [0, 2] Hz, [2,5] Hz

Answer: 16J and 24J, respectively

PSD

Infinite energy signals, such as realizations of stationary random processes or deterministic periodic signals

The total power is obtained as proportional to the integral of density power function S the band

Deterministic or random signals?

Continuous or discrete-time signals?

Energy or power signals?

Real or complex-valued signals?

Time	Ind. var.	Definition	Main property
Continuous-time	f (Hz)	$S(f) = F_f\{R(\tau)\}$	$P = \int_{-\infty}^{\infty} S(f)\mathrm{d}f$
Continuous-time	ω (rad/s)	$S(\omega) = F_\omega\{R(\tau)\}$	$P = \frac{1}{2\pi}\int_{-\infty}^{\infty} S(\omega)\mathrm{d}\omega$
Discrete-time	Ω (rad)	$S(e^{j\Omega}) = F\{R[\ell]\}$	$P = \frac{1}{2\pi}\int_{<2\pi>} S(e^{j\Omega})\mathrm{d}\Omega$

PSD – multiple definitions

$$S(f) = \mathcal{F}\{R(\tau)\}$$

From autocorrelation

$$S(f) = \lim_{T\to\infty} \frac{\mathbb{E}[|X_T(f)|^2]}{T}$$

Expected power distribution from a window of length T

$$S(f) = \lim_{T\to\infty} \frac{|X_T(f)|^2}{T}$$

ESD normalized by time

$$S(e^{j\Omega}) = \mathcal{F}\{R[l]\}$$

$$S(e^{j\Omega}) = \lim_{N\to\infty} \frac{1}{N} \mathbb{E}\left[\left|X_N(e^{j\Omega})\right|^2\right]$$

In practice, sometimes only a single realization is present

Samples are limited, due to, for example, the time over which a process is considered stationary

Special Case: Periodic Signals

Periodic Signals have Fourier Series, thus the PSD has impulses at the Fourier coefficients

$$x(t) = A\cos(2\pi f_c t)$$

$$S(f) = \frac{A^2}{4}\delta(f + f_c) + \frac{A^2}{4}\delta(f - f_c)$$

$$x[n] = A\cos(\Omega_1 n)$$

Where $\frac{\Omega_1}{2\pi} = \frac{m}{N}$

$$S\left(e^{j\Omega}\right) = \frac{\pi A^2}{2}\sum_{p=-\infty}^{\infty}[\delta(\Omega + \Omega_1 + p2\pi) + \delta(\Omega - \Omega_1 + p2\pi)]$$

Mean-square (MS) spectrum or power spectrum

$$\hat{S}_{\mathrm{ms}}[k] = \left|\frac{\mathrm{FFT}\{x_N[n]\}}{N}\right|^2$$

Unlike the PSD, the MS spectrum directly represents the average power (in Watts) of individual sinusoidal components, eliminating the need for integration over a frequency range

The signal power is:

$$\sum_{k=0}^{N_0-1} S_{ms}[k] = \mathcal{P}$$

$$\sum_{k=0}^{N-1} \hat{S}_{ms}[k] = \mathcal{P}$$

$$S_{ms}[k] = |DTFS\{x[n]\}|^2$$

Squared magnitude of the DTFS coefficients

$$\hat{S}_{ms}[k] = \left|\frac{FFT\{x_N[n]\}}{N}\right|^2$$

Estimated MS using a windowed version of x[n]

Thank you!

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